Caution or Constraint? Testing Explanations of a Transformer Risk Profiler Against Exact Bayesian Ground Truth

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Abstract

Robo-advisers and AI assistants increasingly infer clients' attitudes to risk from their behaviour rather than from conversation. When a client declines a financial risk, the adviser must determine whether the client is *cautious*, a preference to respect, or *constrained*, lacking the savings and other buffers that would make a loss affordable. UK and EU suitability rules require the two to be assessed separately. However, field data cannot show whether a model can tell them apart, or whether its explanation of a profile can be trusted, because a client's true traits are never observed. This research builds a testbed in which they are. First, simulated households with known caution and constraint make 24 risky choices under a structural model of household risk-taking, for which the exact Bayesian posterior over both traits, and the exact information each choice contributes to it, can be computed. A set transformer is then trained on these simulations to infer both traits from the choices, and four explanation methods are scored against the exact per-choice information. Across three training seeds, the transformer estimates caution to within an RMSE of 0.06 of the exact posterior mean (prior s.d. 0.40), with calibrated intervals. Raw attention is a weak explanation (Spearman 0.35–0.61 across layers and seeds), attention rollout is better (0.65), integrated gradients varies across equally accurate retrained models (0.30–0.62), and occlusion is the most faithful and most stable (0.76 ± 0.01). Both inference and explanation are bounded by legibility: 24 choices from a low-income household carry about three times the information in the demographic prior, but from a high-income household about a tenth. Explanations of a risk profile are therefore meaningful only where the data identify it, and an aligned adviser should elicit constraints rather than infer them.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Applied computing** → *Economics*.

Keywords

explainability, attention, faithfulness, set transformer, amortised inference, robo-advice, suitability, risk profiling

1 Introduction

Advisers who recommend investments must establish two different things about a client: how much risk the client is willing to take, and how much loss the client can afford to bear. UK rules require recommendations to be suitable for the client's risk tolerance and ability to bear losses [16], and the EU's suitability guidelines draw the same line [15]. The distinction matters because the two call for opposite responses. A client who declines a profitable gamble out of *caution* holds a preference the adviser should respect; a client who declines because the loss would breach rent money is *constrained*, a problem that savings, insurance or credit could relieve. Industry practice has long struggled here: the UK regulator's review of suitability found firms that did not consider a customer's capacity for loss, and issued guidance that it be assessed as a concept distinct from attitude to risk [17].

The distinction is becoming harder to draw. As robo-advisers and AI assistants profile clients at scale [8, 11], more of this judgement is read from behaviour rather than asked about, and behaviour is ambiguous: turning down a gamble is what both a cautious and a constrained client would do. A profile read from behaviour must also be justified in the client's suitability file. For a transformer the obvious justification is its attention weights, which show the choices the model attended to most, but whether such weights explain a model's output is contested [26, 40]. For a real risk profiler neither the profile nor its justification can be checked, since no one observes a client's true caution or constraint.

We therefore construct a setting in which the truth is known. We simulate households whose caution and constraint we set ourselves, using a structural model of household risk-taking [20], and let each household accept or decline 24 gambles. Because each household has only two unknown traits, we can compute exactly what an ideal statistician would conclude from its choices: the *Bayesian posterior*, which gives the probability of every combination of trait values in light of the demographics and the choices observed. We can also compute exactly how much each individual choice moved that conclusion. This second quantity is a ground truth for explanations: a faithful explanation should point to the choices that actually carried the information. Explanation methods are usually evaluated with indirect proxies [2, 24], and ground truth of this kind is rare outside synthetic image benchmarks [5]. Against this reference we ask how close a trained transformer comes to the ideal, and whether standard explanation methods recover the right evidence. Our contributions are:

(1) A testbed for risk profiling in which both the correct profile and the correct explanation are known exactly (Section 3).
(2) Evidence that a 155k-parameter transformer trained only on simulations comes close to the exact Bayesian answer, with calibrated uncertainty, across retraining (Section 5).
(3) A legibility result: choices separate caution from constraint only for households whose constraints bind, which limits both inference and explanation (Section 6).

(4) A comparison of four explanation methods against the exact answer, showing that attention weights are a weak guide and that explanations can change between equally accurate models (Section 7).

Each technical step below is first described in words, so the equations can be skipped on a first reading.

2 Related work

Measuring willingness and ability to take risk. Advisers usually assess risk tolerance with questionnaires, such as the validated instrument of Grable and Lytton [19]. Economists instead elicit risk attitudes from choices between lotteries, as in the Holt–Laury price list [23], and large surveys show that a single self-reported willingness to take risks predicts behaviour across domains [12]. Both traditions measure *willingness*; regulators treat *capacity* for loss as a separate assessment of the client's financial position [15–17]. The two are hard to disentangle in behaviour. Poverty may itself shape risk attitudes [22], while within-person changes in financial slack around payday leave measured risk-taking unchanged [9]. A behavioural profile can therefore attribute to caution what is really constraint, or the reverse; our testbed makes this confusion measurable.

Robo-advice and learning preferences from behaviour. Robo-advice can improve the diversification of under-diversified investors, though it does little for those already diversified [11]. A growing literature lets them learn a client's risk aversion from the client's own decisions, either from observed portfolio choices [3] or through repeated interaction [8]. These models treat risk aversion as a single quantity. More generally, inferring what a person wants from what they do is the problem of inverse reinforcement learning and of assistance games in AI alignment [21, 32]. A central difficulty is identification: without assumptions about how a person decides, behaviour is consistent with many different preferences [4]. We fix those assumptions with a structural model and ask a sharper version of the question: for which clients do their choices distinguish preference from constraint at all?

Neural amortised inference. When a simulator of the data is available but its likelihood is costly to evaluate, a neural network trained on simulated examples can learn to output the posterior directly; this is known as simulation-based or *amortised* inference, since the cost of training is paid once and then shared across all future cases [10, 35]. Transformers [39] trained this way approximate Bayesian posterior predictive distributions [31] and learn simple function classes from examples given in context [18]. When the observations are unordered, permutation-invariant architectures such as DeepSets [41] and the set transformer [29] are the natural choice. These works judge success by accuracy; we also ask whether the network's *reasons* match the exact ones.

Explanations and their faithfulness. An explanation is *faithful* if it reflects the reasoning the model actually used, which is distinct from being plausible to a human [25]. Whether attention weights are faithful is disputed: they can be altered without changing predictions [26], though not always freely [40]. Attention rollout combines attention across layers to improve on raw weights [1]. Alternatives attribute a prediction to inputs through gradients, as in integrated gradients [37], through perturbation, as in occlusion [42], or through Shapley values [30], which are now used to explain credit-risk models [7]. Evaluating any of these is hard without ground truth. Common proxies, such as retraining after removing the features a method ranks highest [24] or checking that explanations change when the model is randomised [2], are indirect, and different methods often disagree on the same prediction [28]. Ground-truth benchmarks exist mainly for images [5]. In finance the question is regulatory as well as scientific: the UK's Prudential Regulation Authority expects banks with internal-model approval to validate their models independently [34], and creditworthiness assessment is classed as high-risk under the EU AI Act [14]. To our knowledge, this is the first test of explanation methods against an exact Bayesian ground truth in a financial decision-making task.

3 Testbed

Overview. Each simulated household is described by its demographics (income band, age and gender), which an adviser would know, and by two hidden traits that the adviser wants to learn. The household is offered 24 gambles, each of the form "risk losing a stake for a chance of winning a larger prize", and its probability of accepting each one depends on both hidden traits. The adviser sees only the demographics and the 24 accept-or-decline decisions. We describe how the probability of acceptance is computed (the simulator), how the hidden traits enter it, and how we compute the exact answer.

Simulator. The structural model [20] borrows an idea from cricket. The Duckworth–Lewis–Stern method, used to reset targets in interrupted matches, treats a batting side's remaining scoring potential as depending jointly on overs left and wickets in hand [13, 36]. By analogy, a household's capacity to absorb a loss depends jointly on its remaining working life (the overs) and its financial buffers (the wickets): a loss that a household with deep buffers and decades of work ahead can absorb may be ruinous for one with neither. A household holds eight buffers $c \in \mathbb{R}^8_{\geq 0}$, measured in months of income: liquid savings, formal credit, insurance, physical assets, a second earner, social network, employability and health. Equation (1) combines them into a single buffer depth w, with diminishing returns to each, and converts depth and the remaining horizon $h = \max(0, 75 - \text{age})$ into a resource level R:

$$w(c) = \sum_{k=1}^{8} \ln\Big(1 + \frac{c_k}{1.5}\Big), \qquad R(h, w) = A\, w^{0.75}\big[1 - e^{-b(w)h}\big], \quad (1)$$

where $b(w)$ falls from 0.085 to 0.030 as w grows, so that a household with thin buffers gains little from a long horizon. A financial loss depletes the five spendable buffers in order of liquidity, while a non-monetary loss destroys a share of a named buffer, here the social network.

A gamble is then judged by how it changes these resources. A gamble pays g with probability p and costs ℓ otherwise, and its value is

$$V = W(p)\frac{\Delta R^{+}}{R_0} - \lambda\, W(1-p)\Big[\frac{\Delta R^{-}}{R_0} + \kappa\, \Delta P_{\text{ruin}}\Big], \quad (2)$$

where R_0 is current resources and $\Delta R^{\pm}$ the change in resources if the gamble is won or lost. This is a form of cumulative prospect theory

[27, 38], the standard descriptive model of choice under risk, applied to resources rather than money. Three features matter. Losses count λ times as much as equal gains (*loss aversion*). Probabilities are distorted by the Prelec weighting function $W(p) = \exp(-(-\ln p)^{0.65})$ [33], which overweights unlikely outcomes. And a penalty $\kappa = 0.55$ applies to any rise in the probability of falling below subsistence, $P_{\text{ruin}} = \exp(-r(c))$, where $r(c)$ is the cash a household could raise in an emergency, including from its social network. Younger households evaluate losses at a partially recovered position, since they have longer to rebuild. Choices are noisy: the household accepts with probability $\sigma((V - 0.012)/0.10)$, where σ is the logistic function, so a gamble of value near 0.012 is accepted about half the time. Remaining parameters follow [20].

Households and hidden traits. Each household has an income band (low, middle or high, with probabilities 0.34, 0.45 and 0.21), an age uniform on 18–65 and a gender, which together fix its expected buffers c and loss aversion λ. We add two hidden traits, drawn independently as $u, v \sim \mathcal{N}(0, 0.4^2)$. *Caution* u scales loss aversion to λe^u, so $u > 0$ means the household dislikes losses more than its demographics predict. *Constraint* v scales every buffer to $c\, e^v$, so $v < 0$ means thinner buffers than its demographics predict. A prior standard deviation of 0.4 means a typical household differs from its demographic profile by a factor of about 1.5 in each trait.

Choices. The stake ℓ is log-uniform on $[0.25, 6]$ months of income, the prize is $m\ell$ with m log-uniform on $[1.2, 10]$, and the success probability is uniform on $[0.25, 0.85]$. With probability 0.35 the loss also destroys a fraction (uniform on $[0.3, 0.9]$) of the household's social network, a non-monetary cost such as strained relations with the family and friends who would otherwise help in an emergency. Each decision is drawn from the simulator's acceptance probability. We simulate 30,000 training, 2,000 validation and 1,500 test households.

Exact posterior. Bayes' rule combines what the demographics imply about the traits (the prior π) with what each choice implies (its likelihood). Because there are only two unknowns, we can evaluate this on a grid. For the first 300 test households we compute the log-likelihood $L_j(u, v)$ of each observed choice j on a 17×17 grid over $[-1.2, 1.2]^2$ and form $p(u, v \mid y_{1:K}) \propto \pi(u, v) \prod_{j \le K} e^{L_j(u,v)}$, where y_j is the jth decision. The mean of this posterior is the best possible estimate of the traits, in the sense of smallest expected squared error.

Exact attribution. To measure how much choice j mattered, we recompute the posterior without it and measure how far the posterior moves. The distance is the Kullback–Leibler (KL) divergence, the standard measure of how different two probability distributions are; it is zero when they are identical and is measured in nats:

$$I_j = \mathrm{KL}\big(p(u, v \mid y_{1:24}) \,\big\|\, p(u, v \mid y_{1:24 \setminus j})\big). \tag{3}$$

On average the six most informative choices carry 78% of the total, so the information is concentrated in a few choices but not trivially so.

4 Model

The exact calculation evaluates the simulator for every candidate trait value, which is slow and becomes infeasible as the number of traits grows. Instead we train a network that maps a household's demographics and choices directly to an estimate of its traits. We use a transformer [39], which processes a list of items called *tokens* and lets each token gather information from the others through *attention weights*. These weights are non-negative and sum to one, which is why they are tempting to read as a statement of which inputs mattered.

Since the order in which a household's choices are observed carries no information, we omit the positional encoding that transformers normally use to track order, which makes the network's output invariant to reordering the choices [29, 41]. Each choice becomes a token built from six features (log stake, log prize multiple, success probability, social-damage fraction, a social-damage flag and the decision itself) by a two-layer embedding. Two further tokens are prepended: a learned summary token (CLS), whose final state is read out as the answer, and a *demographic* token encoding income band, age and gender, which supplies the prior. The network has three layers (width $d = 64$, four attention heads, feed-forward width 256, layer normalisation before each block; 155k parameters) and outputs a Gaussian estimate for each trait, a mean and a standard deviation $(\mu_u, \mu_v, \sigma_u, \sigma_v)$.

Training minimises the Gaussian negative log-likelihood of the true (u, v), which penalises both confident errors and needless uncertainty, so the network learns to report how sure it is. For each training example the number of visible choices is drawn uniformly from 1 to 24 and the rest are hidden, so one network serves every number of choices. We use AdamW with warm-up and cosine decay (peak learning rate 10^{-3}, batch 256, 4,000 steps) in JAX [6], keeping the checkpoint with the best validation loss; one run takes about 11 minutes on a single CPU core. The baseline is DeepSets [41] (44k parameters), which averages per-choice features and so must weight every choice equally; it is trained identically. Each network is trained with three random seeds. Code is available at https://github.com/RISKMASTER16/caution-constraint.

5 Inference quality

We measure accuracy by root mean squared error (RMSE) against the true traits, and closeness to the ideal by the RMSE between a network's estimate and the exact posterior mean. Table 1 and Fig. 1 compare both networks with the exact posterior on the 300 households where it is computed. After 24 choices the transformer's RMSE is 0.348 for caution and 0.385 for constraint, against 0.341 and 0.371 for the exact posterior and 0.412 and 0.419 for a guess from demographics alone. More directly, its estimate differs from the exact posterior mean by an RMSE of 0.06 (caution) and 0.08 (constraint). The advantage of attention over averaging is clearest with little evidence: after four choices the transformer is 0.042 from the exact mean for caution, against 0.074 for DeepSets. With 24 choices the two are within seed variation for caution, and the transformer stays closer for constraint (Table 1). Both networks are calibrated: on the full test set, the transformer's 90% intervals contain the true value in 89% of cases.

The gap that remains is not a failure of the network. Even the exact posterior improves on the demographic guess by only 0.071 (caution) and 0.048 (constraint) in RMSE after 24 choices, and the two traits become entangled: higher caution and thinner buffers

Table 1: Inference quality on 300 test households with an exact posterior. Network entries are mean ± s.d. over three seeds. "Dist." is the RMSE between a network's posterior mean and the Bayes-optimal mean.

K	Method	RMSE u	RMSE v	Dist. u	Dist. v
4	DeepSets	0.392 ± 0.003	0.418 ± 0.002	0.074 ± 0.004	0.076 ± 0.001
	Transformer	0.388 ± 0.002	0.411 ± 0.003	0.042 ± 0.003	0.049 ± 0.007
	Bayes-optimal	0.385	0.406	0	0
24	DeepSets	0.350 ± 0.001	0.384 ± 0.001	0.064 ± 0.002	0.092 ± 0.001
	Transformer	0.348 ± 0.005	0.385 ± 0.000	0.060 ± 0.006	0.078 ± 0.004
	Bayes-optimal	0.341	0.371	0	0
–	Prior only	0.412	0.419		

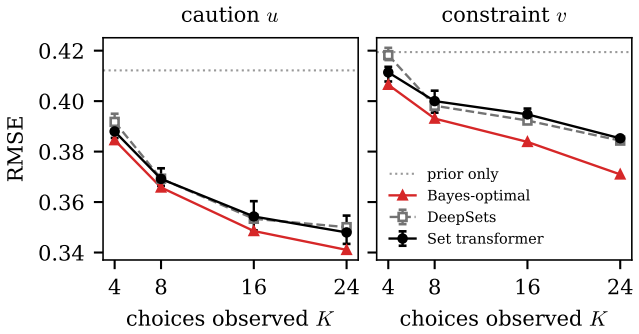


Figure 1: Estimation error against the number of observed choices. Networks: mean over three seeds, bars show the range. The dotted line is the prior.

both lower acceptance, so the average posterior correlation between u and v rises from 0.08 at $K = 4$ to 0.24 at $K = 24$. Observing more choices increasingly pins down a combination of the two traits rather than each separately, which is exactly the distinction suitability rules ask advisers to draw.

6 Legibility

How much can choices reveal at all? We measure this with *Fisher information*: how sharply a household's acceptance probabilities change when a trait changes. The more sharply they change, the more each choice reveals. Information adds across independent choices, and can be compared with the information already in the demographics, the prior precision $1/0.4^2 = 6.25$. Figure 2(a) reports the expected information in 24 choices, evaluated at each household's true traits. For low-income households it is 19.1 for caution and 16.7 for constraint, about three times the prior precision. For middle-income households it is about 2.9, and for high-income households 0.60 and 0.68. Equivalently, a high-income household would need roughly 250 such choices to reveal as much about its caution as its demographics already do; a low-income household needs about eight. Error follows information: at $K = 24$ the exact posterior's RMSE for low-income households is 0.261 (caution) and 0.287 (constraint), while for high-income households it stays close to the prior (0.389 and 0.447). Well-buffered households rarely approach their ruin point on these stakes, so their acceptance probabilities sit near one half and their choices say little about either trait.

Table 2: Faithfulness of explanations against the exact per-choice information (300 households, $K = 24$). Mean ± s.d. over three seeds. Top-6 is the overlap with the six most informative choices; chance is 0.25.

		Spearman with I_j				
Model	Method	All	Low	Mid	High	Top-6
Transformer	Attention, layer 1	0.49 ± 0.09	0.27	0.63	0.60	0.54
	Attention, layer 2	0.43 ± 0.01	0.36	0.52	0.38	0.54
	Attention, layer 3	0.43 ± 0.06	0.33	0.49	0.47	0.47
	Attention rollout	0.65 ± 0.06	0.43	0.80	0.73	0.66
	Integrated gradients	0.48 ± 0.14	0.42	0.57	0.41	0.59
	Occlusion	0.76 ± 0.01	0.70	0.84	0.74	0.71
DeepSets	Integrated gradients	0.56 ± 0.07	0.45	0.71	0.46	0.63
	Occlusion	0.79 ± 0.00	0.77	0.86	0.71	0.76

7 Explanation faithfulness

An adviser that relies on the transformer must be able to say which of a client's choices drove the assessment. Each method below assigns a score to each of a household's 24 choices; we compare the scores with the exact information I_j of Eq. (3).

- *Attention*: the weight the CLS token places on each choice, averaged over heads, in each layer. This is what an explanation dashboard would typically display.
- *Attention rollout* [1]: combines the attention of all three layers, allowing for the direct (residual) path by which each token also passes its own state forward, $\hat{A} = \frac{1}{2}A + \frac{1}{2}I$.
- *Integrated gradients* [37]: accumulates the gradient of the estimates μ_u, μ_v along a straight path from a neutral reference input (the average token, 32 steps) to the actual one, attributing the change in output to each token; summed in absolute value.
- *Occlusion* [42]: deletes one choice and measures how far the network's posterior moves, in KL. It asks the network the same question that Eq. (3) asks the exact posterior.

Integrated gradients and occlusion also apply to DeepSets. For each household we compute the Spearman rank correlation between a method's scores and I_j, which is 1 when the method orders the choices exactly as the truth does and 0 when the orders are unrelated. We also report the overlap between the six choices a method ranks highest and the six most informative (chance = 0.25).

Table 2 and Fig. 2(b) give the results. Raw attention is a weak explanation. Across the three layers and three seeds its correlation with the exact information ranges from 0.35 to 0.61, no layer is consistently best, and in the last layer the six choices it weights most overlap the six most informative at a rate of only 0.47, against 0.25 by chance. Propagating attention through the residual stream helps: rollout reaches 0.65. Integrated gradients averages 0.48 but is unstable: across three retrained transformers whose RMSE differs by at most 0.011, its faithfulness ranges from 0.30 to 0.62. This is a form of the disagreement between explanations that practitioners report [28], here between equally accurate models rather than between methods. Occlusion is both the most faithful and the most stable, at

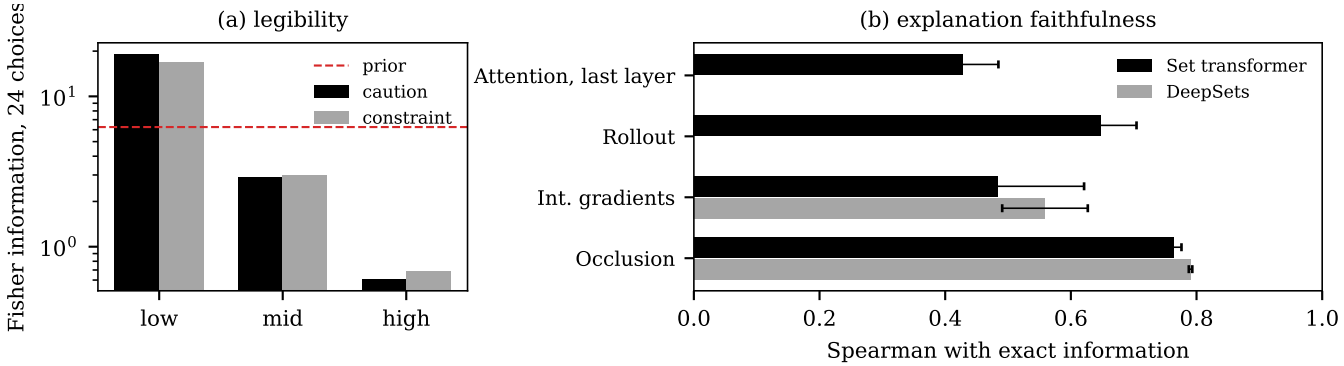


Figure 2: (a) Expected Fisher information in 24 choices by income band, against the prior precision (log scale). (b) Within-household Spearman correlation between each attribution method and the exact per-choice information I_j; mean ± s.d. over three seeds.

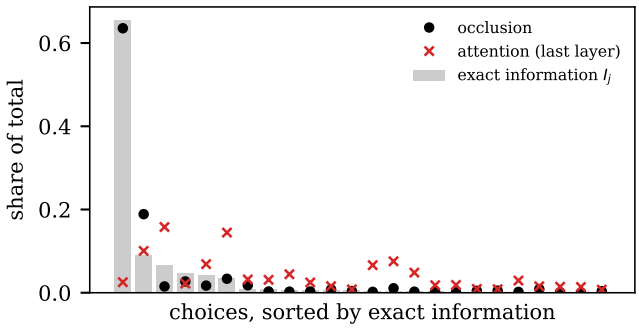


Figure 3: One household's 24 choices, sorted by their exact information I_j (bars), with each method's normalised attribution. The household is the low-income one with the median occlusion faithfulness in its band.

0.76 ± 0.01 with a top-six overlap of 0.71. DeepSets, which has no attention to inspect, is explained about as faithfully by occlusion (0.79), so faithfulness here is a property of the explanation method rather than of the architecture.

Faithfulness also varies by income band, and not in the direction one might hope. Every method is most faithful for middle-income households, and six of the eight method–model pairs are least faithful for low-income households (transformer occlusion 0.70, against 0.84 for the middle band), although their choices carry the most information: 0.61 nats of total leave-one-out information, against 0.05 for high-income households. A plausible reason is that near the ruin point each choice's contribution depends on the others, which single-choice attributions cannot represent. For high-income households a ranking can be faithful yet explain almost nothing, since their 24 choices together carry less than a tenth of the information of a low-income household's.

Figure 3 shows, for a household selected by rule rather than by hand, why a rank correlation can understate the problem. A single choice, accepting a long-shot gamble (success probability 0.27) whose loss would also damage the household's social network, carries 65% of its exact information. Occlusion gives that choice 64% of its weight; last-layer attention gives it 3%, and puts its largest weight on a choice worth 7%. Attention still ranks this household's choices fairly well (Spearman 0.67), so a rank metric alone would pass it, yet an adviser shown these weights would miss the single most revealing decision.

8 Discussion

Three lessons follow for AI systems that profile clients' risk. *Do not read attention as justification, and validate explanations, not only accuracy.* The weights a transformer displays are among the least faithful explanations we tested: a suitability file that cited the last layer's top choices would cite the wrong evidence about half the time. Explanations can also differ between models that are equally accurate, as integrated gradients does here, so model validation of the kind supervisors expect [34] should test explanations as well as accuracy. Occlusion is the most faithful and the most stable and costs one forward pass per choice; rollout is nearly free and better than raw weights. *Report uncertainty, not only a profile.* For most households the honest posterior over caution is close to the prior; a point estimate would present a guess as a measurement, and an explanation of it would explain the prior. *Ask rather than infer.* Because constraint is legible from behaviour only where it binds, an aligned adviser should elicit buffers directly, as the ability-to-bear-losses assessment intends, and reserve behavioural inference for caution. A similar distinction, between a borrower's willingness and ability to repay, arises in creditworthiness assessment, which the EU AI Act classifies as high-risk [14]; a testbed with exact ground truth offers one way to validate explanations before deployment.

Limitations. The simulator is one structural model with assumed parameters, including a choice-noise scale that largely determines how uninformative well-off households' choices are; the legibility gradient should be re-examined under estimated noise. The exact reference covers 300 households, with 65 in the high-income band. Gambles are drawn at random, whereas real advisers observe richer, non-random behaviour. Integrated gradients depends on its baseline, and we tested one. A natural extension is active elicitation: letting the transformer choose the question that maximises expected information.

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